

Who Gets Constituent Service?

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These plots can be found in the CorRep/figs folder on dropbox and the APSA2020/figs folder on github.	

```
source(here::here("setup.R"))

library(ggplot2); theme_set(theme_bw())
options(
  ggplot2.continuous.color = "viridis",
  ggplot2.continuous.fill = "viridis"
)
scale_color_discrete <- function(...)
  scale_color_viridis_d(...)
scale_fill_discrete <- function(...)
  scale_fill_viridis_d(...)

load(here::here("data/all_contacts.Rdata"))
d <- all_contacts

d %<>%
  mutate(TYPE = replace_na(TYPE, "NA"),
         constituent = ifelse(TYPE == 1, "Indiv. Constituent Service", "Other")) %>%
  add_count(icpsr, constituent, agency, year, name = "n_per_year_agency_type") %>%
  add_count(icpsr, year, name = "n_per_year") %>%
  mutate(percent_type_to_agency = (n_per_year_agency_type/n_per_year)*100)

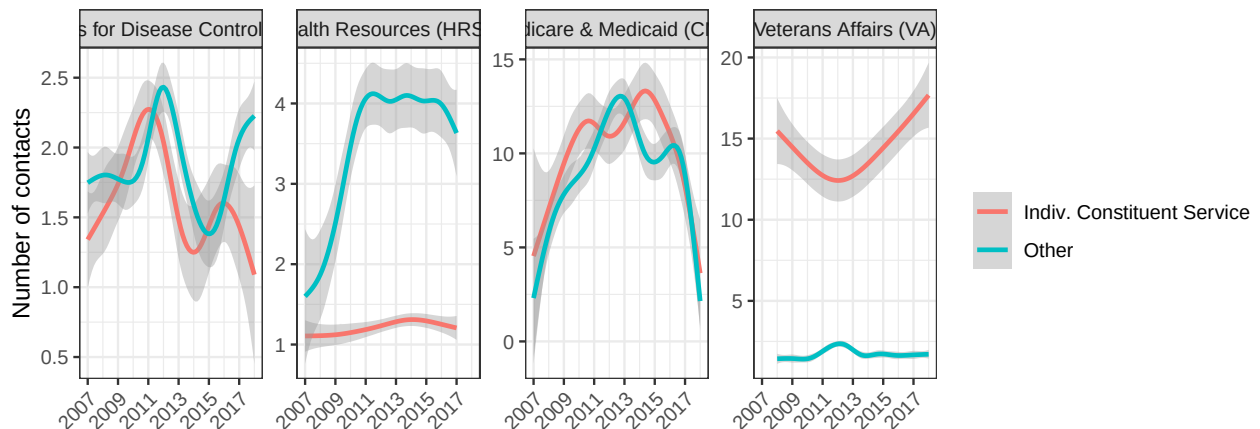
d1 <- d %>%
  filter(year > 2006, year <2020) %>%
  select(icpsr, year, agency, constituent, n_per_year_agency_type, percent_type_to_agency, party_name) %>%
  filter(agency %in% c("VA", #"SSA",
                     "DHHS_CMS", "DHHS_CDC", "DHHS_HRSA")) %>%
  distinct()

d1$agency %<>%
```

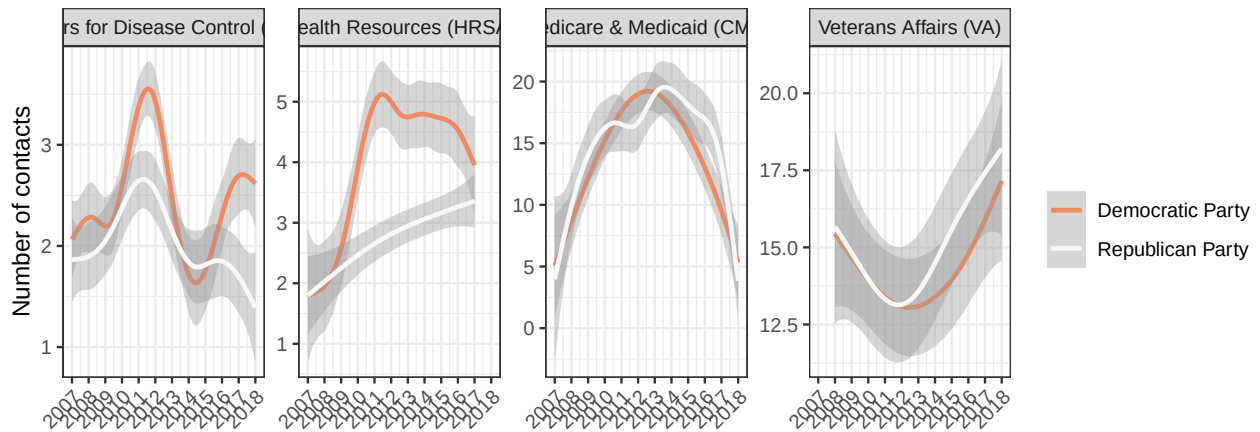
```
str_replace("VA", "Veterans Affairs (VA)") %>%
str_replace("DHHS_CMS", "Medicare & Medicaid (CMS)") %>%
str_replace("DHHS_CDC", "Centers for Disease Control (CDC)") %>%
str_replace("DHHS_HRSA", "Health Resources (HRSA)")
```

Number of legislator contacts per agency per year 2007-2017

```
d1 %>%
  ggplot() +
  aes(x = year, color = constituent, y = n_per_year_agency_type) +
  #geom_line(aes(group= icpsr), alpha = .1) +
  geom_smooth() +
  labs(x = "", y = "Number of contacts", color = "") +
  facet_wrap(. ~ agency, scales = "free_y", nrow = 1) +
  scale_x_continuous(breaks = seq(2007, 2017, 2)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

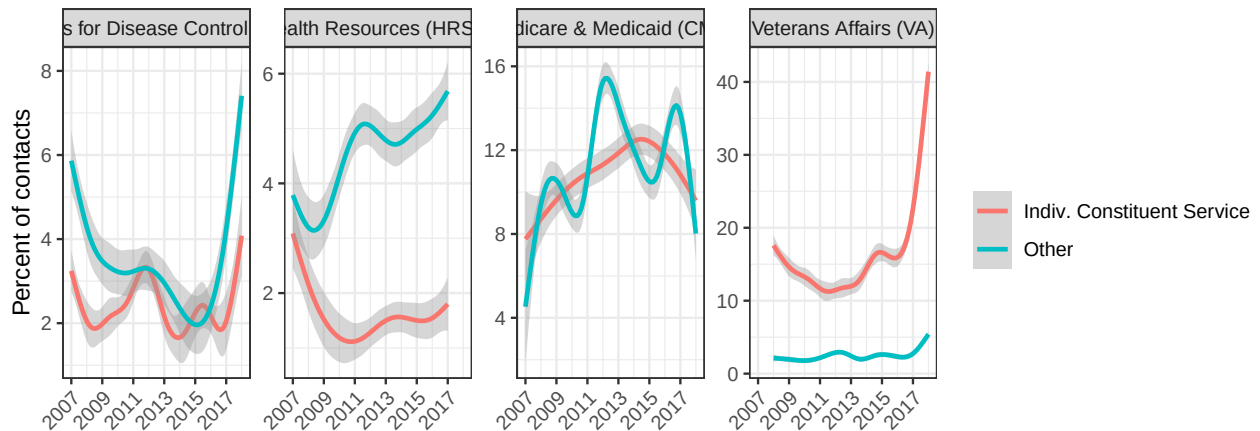


```
d1 %>%
  filter(party_name != "Independent") %>%
  group_by(icpsr, party_name, agency, year) %>%
  summarise(n_per_year_agency = sum(n_per_year_agency_type)) %>%
  ggplot() +
  aes(x = year, color = party_name, y = n_per_year_agency) +
  #geom_line(aes(group= icpsr), alpha = .1) +
  geom_smooth() +
  labs(x = "", y = "Number of contacts", color = "") +
  facet_wrap(. ~ agency, scales = "free_y", nrow = 1) +
  scale_x_continuous(breaks = seq(2007, 2019, 1)) +
  scale_color_brewer(palette = "RdBu") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Percent of legislator contacts per agency per year 2007-2017

```
d1 %>%
  ggplot() +
  aes(x = year, y = percent_type_to_agency, color = constituent) +
  #geom_line(aes(group= icpsr), alpha = .1) +
  geom_smooth() +
  labs(x = "", y = "Percent of contacts", color = "") +
  facet_wrap(~ agency, scales = "free_y", nrow = 1) +
  scale_x_continuous(breaks = seq(2007, 2017, 2)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Constituent type count:

```
load(here::here("data", "dcounts_min.Rdata"))

dcounts_min$agency %<>%
  str_replace("VA", "Veterans Affairs (VA)") %>%
  str_replace("DHHS_CMS", "Medicare & Medicaid (CMS)") %>%
  str_replace("DHHS_CDC", "Centers for Disease Control (CDC)") %>%
  str_replace("DHHS_HRSA", "Health Resources (HRSA)") %>%
  str_replace("Treasury_IRS", "IRS") %>%
  str_replace("NARA", "National Archives (NARA)") %>%
```

```

str_replace("DOL.*", "Labor (DOL VETS, SOL, OWCP, & OFCCP)") %>%
str_replace("Treasury_Fiscal", "Treasury Fiscal Service") %>%
str_replace("SSA", "Social Security (SSA)")

names(dcounts_min) %<>% str_replace("vet", "Veteran") %>%
  str_replace("hardship", "Hardship situation") %>%
  str_replace("lowincome", "Low income") %>%
  str_replace("military", "Military family or veteran") %>%
  str_replace("senior", "Senior")

dcounts_min %>%
  # melt count vars to long
  select(agency, starts_with("per")) %>%
  gather("type", "count", -agency) %>%
  mutate(type = type %>% str_remove(".*_")) %>%
  # count
  group_by(agency, type) %>%
  summarize(count = sum(count)) %>%
  ungroup() %>%
  filter(count > 0) %>% #select(type) %>% distinct()
  filter(type %in% c("Hardship situation",
    # "Military family or veteran",
    "Senior",
    "Veteran",
    "Low income")) %>%
  group_by(type) %>%
  add_count(name = "n_agency") %>% # n agencies per type
  ungroup() %>%
  ggplot() +
  aes(y = count, x = agency, label = count, width = .15*n_agency) +
  geom_col(position = "dodge", preserve = "single") +
  geom_text(hjust = -.1) +
  coord_flip()+
  scale_y_continuous(expand = expand_scale(mult = c(0,.2))) + # pad y axis so there is room for n text
  labs(y = "Number of Constituent Service Letters", #paste("Letters on Behalf", type),
    x = "Agency") +
  facet_wrap("type", scales = "free", ncol = 1) +
  theme_bw() +
  theme(panel.grid.minor.y = element_blank(),
    panel.grid.major.y = element_blank())

```

Constituent type share of total per agency

```

dcounts_min %>%
  #select(-per_icpsr_chamber_year_agency_type) %>%
  group_by(agency) %>%
  summarise(across(starts_with("per_"), sum)) %>%
  gather(key = type, value = count, -agency) %>%
  mutate(type = str_remove_all(type, ".*_") %>% str_replace("type", "All") ) %>%
  group_by(agency) %>%
  mutate(agencycount = sum(count)) %>%
  filter(count > 0, agencycount > count) %>%

```

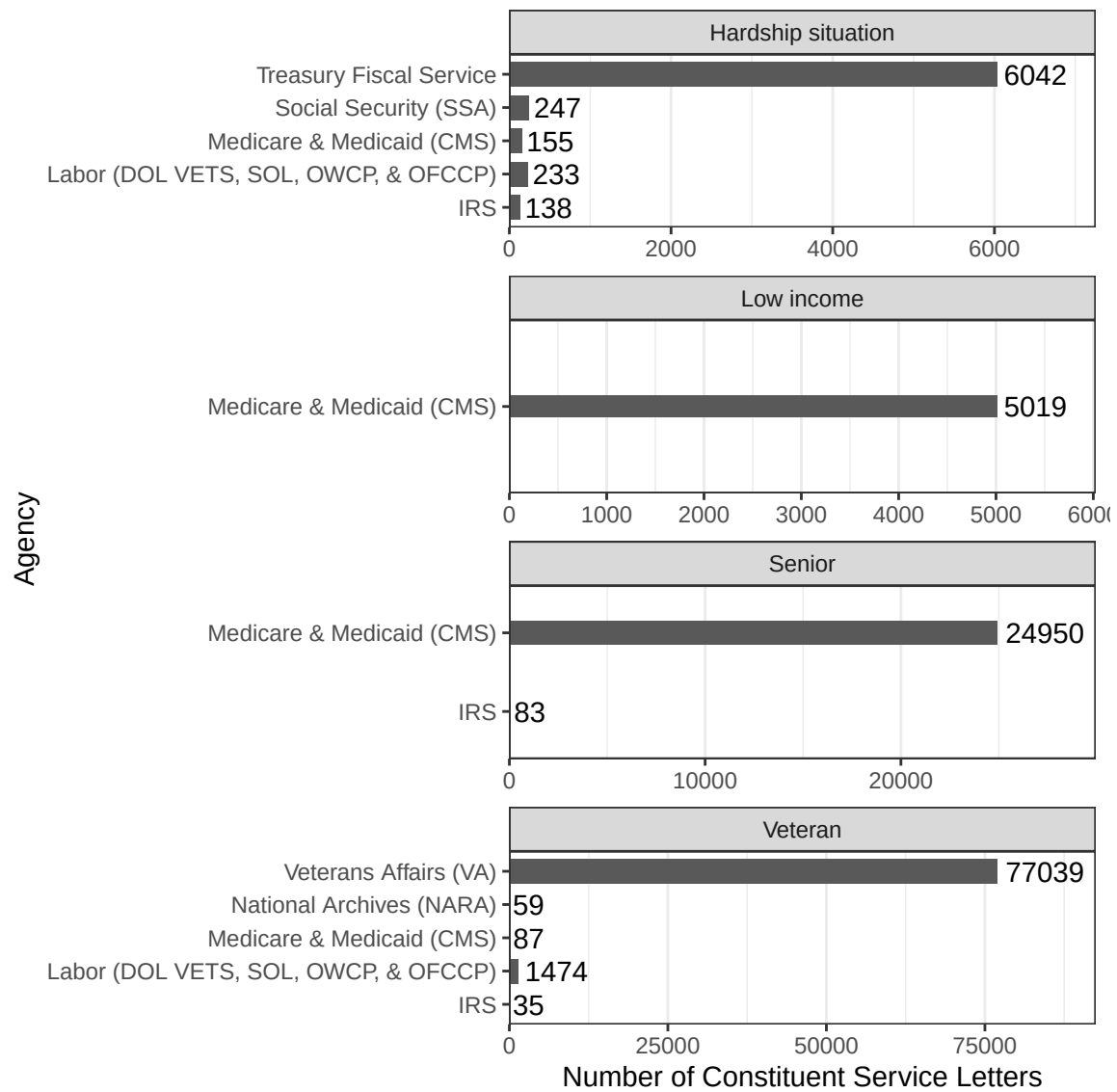


Figure 1: Number of Legislator Letters per Type of Constituent

```

group_by(agency) %>%
add_count(name = "n_types") %>%
ggplot() +
aes(x = agency, y = count, fill = factor(type), label = count, width = .2*n_types) +
geom_col(position = "dodge") +
#geom_text(position = "dodge", hjust = 0) +
coord_flip()+
scale_y_continuous()+#expand = expand_scale(mult = c(0,.2))) +
labs(y = "",
x = "Agency",
fill = "Type of Constituent") +
facet_wrap("agency", scales = "free", ncol = 1) +
theme_bw() +
theme(panel.border = element_blank(),
axis.ticks = element_blank(),
panel.grid.minor.y = element_blank(),
panel.grid.major.y = element_blank(),
strip.background = element_blank(),
strip.text.x = element_blank())

```



Figure 2: Share of Legislator Contacts by Type of Constituent

Member-Congress Percentiles

```
load(here::here("data", "dcounts_mid.Rdata"))

dcounts %<>% filter(chamber != "President")

dcounts %<>% select(chamber, last_name, state_abbrev, party_name, congress, year, agency, icpsr, TYPE, )

# save(dcounts, file = here::here("data" , "dcounts_mid.Rdata"))

dcounts %<>%
  mutate(member_state = str_c(last_name, ", ", str_sub(party_name, 1, 1), "-", state_abbrev))

percentplot <- function(var, title, x){

  dcounts$var <- var

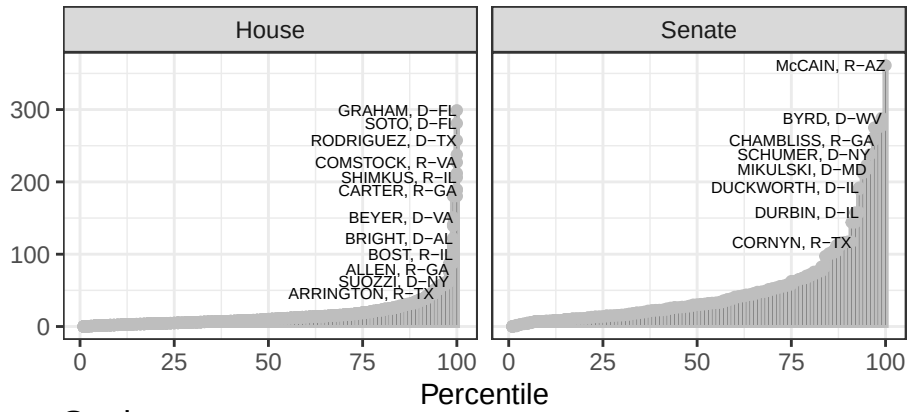
  dcounts %>%
    group_by(member_state, chamber, congress) %>% summarise(n = sum(var)) %>%
    group_by(member_state, chamber) %>% mutate(mean = mean(n)) %>% ungroup() %>%
    group_by(chamber) %>% mutate(Percentile = ntile(mean, 100)) %>%
    arrange(-Percentile) %>%
    dplyr::select(member_state, Percentile, mean, chamber) %>% distinct() %>%
    ggplot() +
      geom_col(aes(x = Percentile, y = mean), color = "grey", position = "dodge") +
      geom_point(aes(x = Percentile, y = mean), color = "grey") +
      geom_text(aes(x = Percentile, y = mean, label = ifelse(Percentile > 90, member_state, "")), check_o
      facet_grid(. ~ chamber , scales = "free_y") + theme_bw() +
      labs(#title = "Average Legislator Requests per Congress by Percentile",
        title = title,
        y = "",
        x = x)
  }

percentplot(dcounts$per_icpsr_chamber_year_agency_vet, "Veterans", x = "Percentile")
#percentplot(dcounts$per_icpsr_chamber_year_agency_military, "Military Families, Personal, and Veterans
percentplot(dcounts$per_icpsr_chamber_year_agency_senior, "Seniors", x = "Percentile")
percentplot(dcounts$per_icpsr_chamber_year_agency_hardship, "Constituents Experiencing Hardship", x = "
percentplot(dcounts$per_icpsr_chamber_year_agency_lowincome, "Low Income Constituents", x = "Percentile")
```

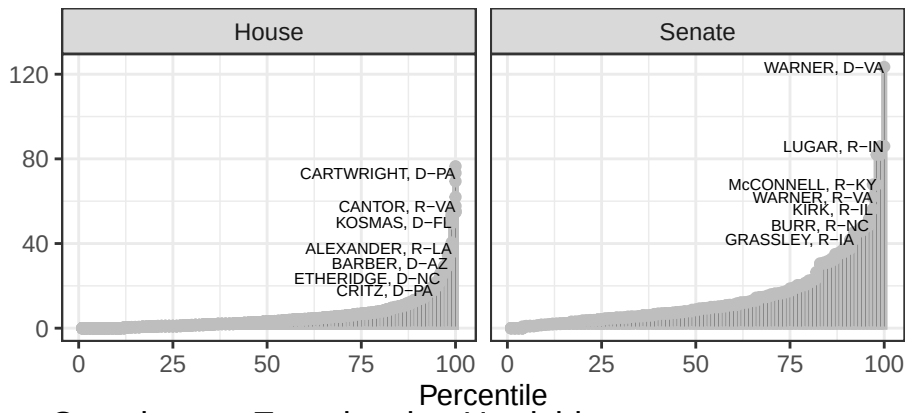
TODO

Maybe also any interesting patterns in the volume of total letters on behalf of constituent types over time? Bars for each Congress broken down by constituent type Basically, this would show how many letters there are on behalf of different constituent types, where the letters are in the agencies, who is writing the letters, and when the letters were written.

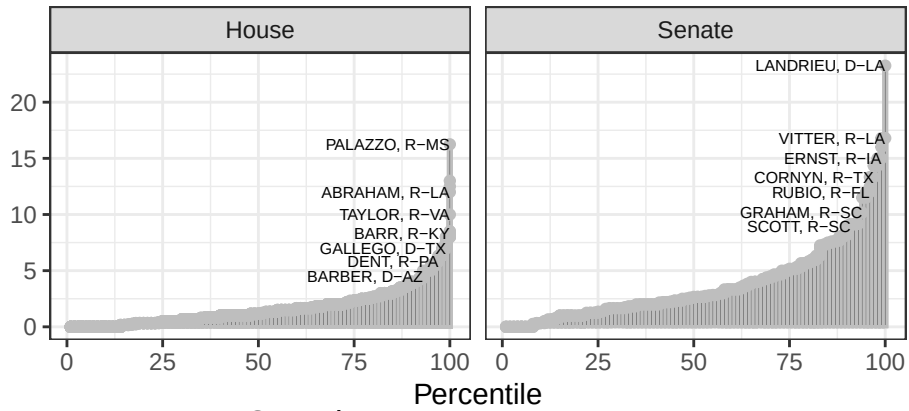
Veterans



Seniors



Constituents Experiencing Hardship



Low Income Constituents

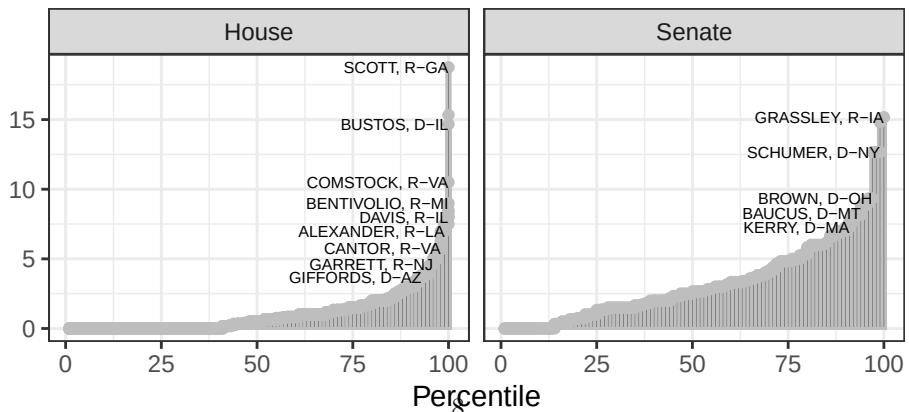


Figure 3: Average Legislator Requests per Congress by Constituent Type